Variance Reduction

STAT 525 - Topics in Computational Statistics

Shulei Wang

University of Illinois at Urbana-Champaign

shuleiw@illinois.edu

https://sites.google.com/view/shulei-wang/



Overview

Finite-Sum Problem

Gradient Aggregation

Implicit Bias

Finite-Sum Problem

$$\min_x f(x) = \frac{1}{n}\sum_{i=1}^{n} f_i(x)$$

- This is a very general problem in machine learning and statistics fields.
  - Logistic regression
  - Lasso
  - Deep neutral network
  - ...
- This problem also appears in distributed optimization, where each computer only have access to one $f_i(x)$.

Gradient Descent

$$x^{(k+1)} = x^{(k)} - t^{(k)} \frac{1}{n} \sum_{i=1}^{n} \nabla f_i(x)$$

- Converge rate:

$$O\left(\frac{\kappa - 1}{\kappa + 1}\right)^k$$

- Iteration cost: $O(n)$
- Overall complexity:

$$O\left(n\kappa \log\left(\frac{1}{\epsilon}\right)\right)$$

- Accelerated gradient descent can achieve $O\left(n\sqrt{\kappa}\log\left(1/\epsilon\right)\right)$

Stochastic Gradient Descent

$$x^{(k+1)} = x^{(k)} - t^{(k)} \nabla f_{i^{(k)}}(x)$$

- Converge rate:

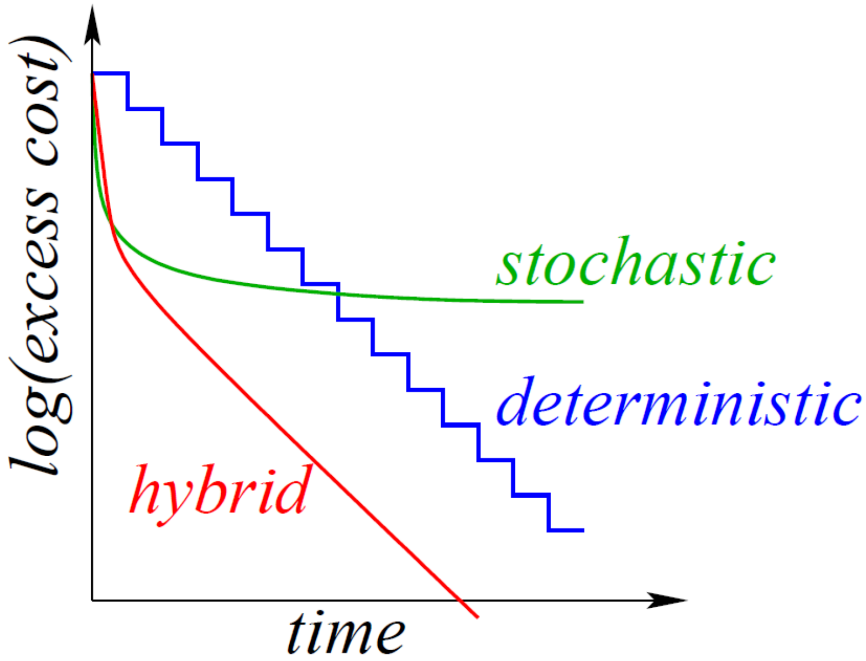
$$O\left(\frac{1}{k}\right)$$

- Iteration cost: $O(1)$
- Overall complexity:

$$O\left(\frac{1}{\epsilon}\right)$$

- Better choice when n is large and ϵ is not small.

GD vs SGD



- Can we have best of both worlds? Can we develop some algorithm that has linear convergence and cheap iteration cost?

Stochastic Gradient Descent

- Recall the stochastic gradient descent is defined as

$$x^{(k+1)} = x^{(k)} - t^{(k)} \nabla f_{i^{(k)}}(x)$$

- If we choose constant step size,

$$\mathbb{E}\left(\|x^{(k)} - x^*\|^2\right) \leq \left(1 - 2tm + t^2L^2\right)^k \|x^{(0)} - x^*\|^2 + \frac{tB^2}{2m - tL^2}.$$

  SGD walks around in a neighborhood of x^* because of noise in gradient ($B > 0$).
- To mitigates oscillation, we choose smaller step size. How about reducing the variance of gradient?

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Control Variates

- Control variates is an idea of variance reduction originated from Monte Carlo simulation. Suppose we know

$$\mathbb{E}(X) = \mu \qquad \text{and} \qquad \mathbb{E}(Y) = 0$$

To estimate μ, we can adopt

$$\hat{\mu}_\alpha = X - \alpha Y.$$

- $\hat{\mu}_\alpha$ is an unbiased estimator. The variance of $\hat{\mu}_\alpha$ is

$$\text{Var}(\hat{\mu}_\alpha) = \text{Var}(X) + \alpha^2 \text{Var}(Y) - 2\alpha \text{Cov}(X, Y)$$

If we choose $\alpha = \text{Cov}(X, Y)/\text{Var}(Y)$, we can know $\text{Var}(\hat{\mu}_\alpha) < \text{Var}(X)$ if $\text{Cov}(X, Y) \neq 0$.

Gradient Aggregation

- We can apply a similar idea of control variates to gradient estimation.
- In stochastic gradient descent, we have an unbiased estimation of gradient

$$X = \nabla f_{i^{(k)}}(x) \qquad \text{and} \qquad \mathbb{E}(X) = \frac{1}{n}\sum_{i=1}^{n} \nabla f_i(x)$$

- To reduce variance, we could find a historical gradient Y which we believe is highly correlated with X. Then, instead of using X, we can use $X - Y$ to estimate gradient. This idea is called gradient aggregation (also called incremental gradient method).

Stochastic Variance Reduced Gradient

- Stochastic variance reduced gradient (SVRG) is one of most popular gradient aggregation method, which is introduced by "Johnson, R. and Zhang, T., 2013. Accelerating stochastic gradient descent using predictive variance reduction. Advances in neural information processing systems, 26, pp.315-323."
- The key idea is that the gradient is estimated by

$$\underbrace{\nabla f_{i^{(k)}}\left(x^{(k)}\right)}_{X} - \underbrace{\left(\nabla f_{i^{(k)}}\left(x^{old}\right) - \frac{1}{n}\sum_{i=1}^{n}\nabla f_i(x^{old})\right)}_{Y}$$

- Y is highly correlated with X when $x^{(k)} \approx x^{old}$. Clearly, $\mathbb{E}(Y) = 0$.

Stochastic Variance Reduced Gradient

- But how should we look for x^{old}? Adopt the epochs to find x^{old}.
- In sth epoch
  1. Take the current iterate as x_s^{old} and compute the batch gradient $\nabla f(x_s^{old})$ (which require computation time n)
  2. We the take a inner loop (m iterations) to update points as stochastic gradient descent. Randomly pick $i^{(k)}$ and then

$$x_s^{(k+1)} = x_s^{(k)} - t^{(k)} \left(\nabla f_{i^{(k)}}(x_s^{(k)}) - \left(\nabla f_{i^{(k)}}(x_s^{old}) - \nabla f(x_s^{old}) \right) \right)$$

- How to update x_s^{old}?
  - Choice I: $x_{s+1}^{old} = x_s^{(m)}$
  - Choice II: $x_{s+1}^{old} = m^{-1} \sum_{k=1}^{m} x_s^{(k)}$
  - Choice III: $x_{s+1}^{old} = x_s^{(k)}$ where k is randomly drawn from $\{1, \ldots, m\}$

Stochastic Variance Reduced Gradient

- Several points you need to be aware of
  - Here, we can choose constant step size $t^{(k)} = t$
  - Batch gradient is computed only once per epoch
  - Each epoch contains $2m + n$ gradient computations
  - In implementation, this is a double iterative method.
  - The required storage of SVRG is small

Convergence Analysis

- Assume each $f_i(x)$ is convex and M-smooth, and $f(x)$ is μ-strongly convex. If we choose m large enough such that

$$\rho = \frac{1}{\mu t(1-2Mt)m} + \frac{2Mt}{1-2Mt} < 1,$$

then

$$\mathbb{E}(f(x_s^{old}) - f(x^*)) \leq \rho^s (f(x_0^{old}) - f(x^*))$$

- If we choose $t = c/M$ for a small constant c and choose $m \geq O(\kappa)$, then we need $O(\log(1/\epsilon))$ epochs to achieve ϵ accuracy and the total computational cost is

$$O\left((n+\kappa)\log(1/\epsilon)\right)$$

which is smaller than $O\left(n\kappa\log(1/\epsilon)\right)$ in GD.

Proof: In this proof, we assume

$$x_{s+1}^{old} = \frac{1}{m} \sum_{k} x_s^{(k)}.$$

Step 1: We can show

$$\frac{1}{n} \sum_{i=1}^{n} \|\nabla f_i(x) - \nabla f_i(x^*)\|^2 \leq 2M(f(x) - f(x^*))$$

- The definition of M-smooth suggests that

$$\frac{1}{2M} \|\nabla f_i(x) - \nabla f_i(x^*)\|^2 \leq f_i(x) - f_i(x^*) - \nabla f_i(x^*)^T (x - x^*)$$

- We can get the conclusion by summing up and noting $\nabla f(x^*) = 0$.

Proof: Step 2: We define the search direction

$$g_s^{(k)} = \nabla f_{i^{(k)}}(x_s^{(k)}) - \left(\nabla f_{i^{(k)}}(x_s^{old}) - \nabla f(x_s^{old})\right)$$

and show

$$\mathbb{E}(\|g_s^{(k)}\|^2) \leq 4M[f(x_s^{(k)}) - f(x^*) + f(x_s^{old}) - f(x^*)]$$

Here $\mathbb{E}$ is taken with respect to $i^{(k)}$ and conditioned on everything else. Clearly, $\mathbb{E}(\|g_s^{(k)}\|^2)$ is small when $x_s^{(k)}$ and x_s^{old} are closer to x^*. The variance is indeed reduced!

Proof: Step 2:

$$\mathbb{E}(\|g_s^{(k)}\|^2) \leq 4M[f(x_s^{(k)}) - f(x^*) + f(x_s^{old}) - f(x^*)]$$

- By definition,

$$\begin{aligned}
&\mathbb{E}(\|g_s^{(k)}\|^2) \\
=&\mathbb{E}\|\nabla f_{i^{(k)}}(x_s^{(k)}) - \nabla f_{i^{(k)}}(x^*) + \nabla f_{i^{(k)}}(x^*) - \nabla f_{i^{(k)}}(x_s^{old}) + \nabla f(x_s^{old})\|^2 \\
\leq&2\mathbb{E}\|\nabla f_{i^{(k)}}(x_s^{(k)}) - \nabla f_{i^{(k)}}(x^*)\|^2 + 2\mathbb{E}\|\nabla f_{i^{(k)}}(x^*) - \nabla f_{i^{(k)}}(x_s^{old}) + \nabla f(x_s^{old})\|^2 \\
\leq&2\mathbb{E}\|\nabla f_{i^{(k)}}(x_s^{(k)}) - \nabla f_{i^{(k)}}(x^*)\|^2 + 2\mathbb{E}\|\nabla f_{i^{(k)}}(x^*) - \nabla f_{i^{(k)}}(x_s^{old})\|^2 \\
\leq&4M[f(x_s^{(k)}) - f(x^*) + f(x_s^{old}) - f(x^*)]
\end{aligned}$$

  Last step, we use results in Step 1.

Proof: Step 3: We can show one update analysis

$$\begin{aligned}
&\mathbb{E}\|x_s^{(k+1)} - x^*\|^2 \\
\leq &\|x_s^{(k)} - x^*\|^2 - 2t(1-2Mt)(f(x_s^{(k)}) - f(x^*)) + 4Mt^2(f(x_s^{old}) - f(x^*))
\end{aligned}$$

Here $\mathbb{E}$ is taken with respect to $i^{(k)}$ and conditioned on the history.

- By convexity of f and unbiasedness of $g_s^{(k)}$,

$$\begin{aligned}
&\mathbb{E}\|x_s^{(k+1)} - x^*\|^2 = \mathbb{E}\|x_s^{(k)} - tg_s^{(k)} - x^*\|^2 \\
=&\|x_s^{(k)} - x^*\|^2 - 2t(x_s^{(k)} - x^*)^T\mathbb{E}(g_s^{(k)}) + t^2\mathbb{E}(g_s^{(k)})^2 \\
\leq&\|x_s^{(k)} - x^*\|^2 - 2t(x_s^{(k)} - x^*)^T\nabla f(x_s^{(k)}) \\
&+ 4t^2M[f(x_s^{(k)}) - f(x^*) + f(x_s^{old}) - f(x^*)] \\
\leq&\|x_s^{(k)} - x^*\|^2 - 2t(f(x_s^{(k)}) - f(x^*)) \\
&+ 4t^2M[f(x_s^{(k)}) - f(x^*) + f(x_s^{old}) - f(x^*)] \\
\leq&\|x_s^{(k)} - x^*\|^2 - 2t(1-2Mt)(f(x_s^{(k)}) - f(x^*)) + 4Mt^2(f(x_s^{old}) - f(x^*))
\end{aligned}$$

Proof: Step 4: We can show one epoch analysis

$$\mathbb{E}(f(x_{s+1}^{old}) - f(x^*)) \leq \rho \mathbb{E}(f(x_s^{old}) - f(x^*)).$$

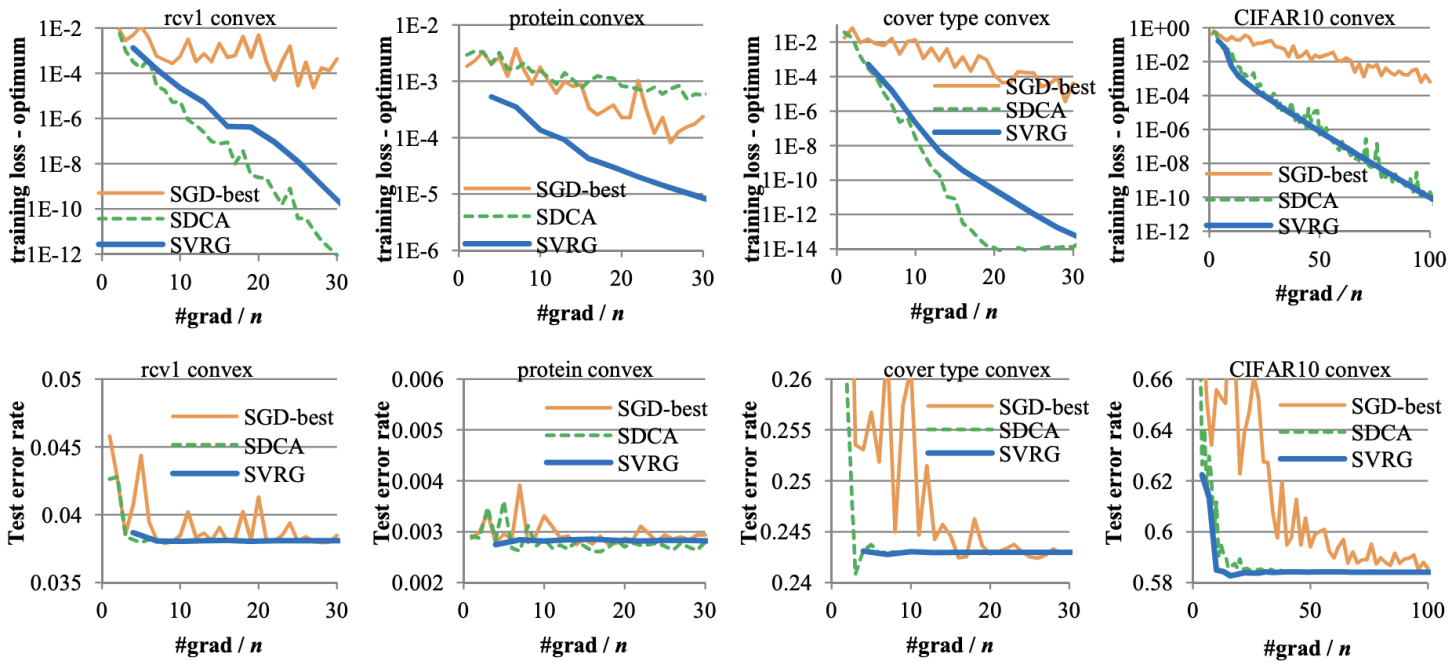
Here $\mathbb{E}$ is taken with respect to all the history.

- By results in Step 3, strong convexity and the definition of x_{s+1}^{old},

$$\begin{aligned}
&2t(1-2Mt)m\mathbb{E}(f(x_{s+1}^{old}) - f(x^*)) \\
\leq &2t(1-2Mt)\sum_{k=0}^{m-1}\mathbb{E}(f(x_s^{(k)}) - f(x^*)) + \mathbb{E}\|x_s^{(m)} - x^*\|^2 \\
\leq &\mathbb{E}\|x_s^{(0)} - x^*\|^2 + 4mMt^2\mathbb{E}(f(x_s^{old}) - f(x^*)) \\
= &\mathbb{E}\|x_s^{old} - x^*\|^2 + 4mMt^2\mathbb{E}(f(x_s^{old}) - f(x^*)) \\
\leq &\frac{2}{\mu}\mathbb{E}(f(x_s^{old}) - f(x^*)) + 4mMt^2\mathbb{E}(f(x_s^{old}) - f(x^*))
\end{aligned}$$

- We can finish the proof by reorganizing these terms.

Numerical Results



- From paper "Johnson, R. and Zhang, T., 2013. Accelerating stochastic gradient descent using predictive variance reduction. Advances in neural information processing systems, 26, pp.315-323."

Extension I of SVRG

- In SVRG, a uniform sampling is adopted, i.e. all the $f_i(x)$ are selected in the same probability. This work well when all $f_i(x)$ are all M-smooth.
- When $f_i(x)$ is M_i-smooth and M_i are very different from each other, we could choose an adaptive sampling strategy

$$\mathbb{P}(i^{(k)} = i) = \frac{M_i}{\sum_i M_i}.$$

  Functions with rapidly change (large M_i) are chosen more often.
- This strategy can also improve total computational cost

$$O\left(\left(n + \frac{M_{max}}{\mu}\right)\log(1/\epsilon)\right) \Rightarrow O\left(\left(n + \frac{M_{avg}}{\mu}\right)\log(1/\epsilon)\right)$$

Extension II of SVRG

- SVRG can also be accelerated by Nesterov's acceleration technique, i.e. incorporating the history information.
- This acceleration can also improve total computational cost

$$O\left(\left(n+\frac{M}{\mu}\right)\log(1/\epsilon)\right) \Rightarrow O\left(\left(n+\sqrt{\frac{Mn}{\mu}}\right)\log(1/\epsilon)\right)$$

- Read details in "Shang, F., Jiao, L., Zhou, K., Cheng, J., Ren, Y. and Jin, Y., 2018, November. Asvrg: Accelerated proximal svrg. In Asian Conference on Machine Learning (pp. 815-830). PMLR." and Chapter 5.4 in First-order and Stochastic Optimization Methods for Machine Learning.

Other Gradient Aggregation Methods

With this similar ideas, there are many other gradient aggregation methods

- Defazio, A., Bach, F. and Lacoste-Julien, S., 2014. SAGA: A fast incremental gradient method with support for non-strongly convex composite objectives. arXiv preprint arXiv:1407.0202.
- Nguyen, L.M., Liu, J., Scheinberg, K. and Takáč, M., 2017, July. SARAH: A novel method for machine learning problems using stochastic recursive gradient. In International Conference on Machine Learning (pp. 2613-2621). PMLR.
- Nguyen, L.M., van Dijk, M., Phan, D.T., Nguyen, P.H., Weng, T.W. and Kalagnanam, J.R., 2019. Finite-sum smooth optimization with sarah. arXiv preprint arXiv:1901.07648.

Overview

Finite-Sum Problem

Gradient Aggregation

Implicit Bias

Role of Optimization Algorithm

$$\min_x f(x)$$

- We only talk about how fast we can find a minimizer x^*. It seems the choice of optimization algorithm only affect speed.
- But the minimizer is sometimes not unique, so different optimization algorithm can help pick different minimizers. This is called implicit bias.
- In a word, optimization algorithm helps select models.

Overparameterized Model: Linear Regression

- We can consider a few examples where the minimizer is not unique. Most examples come from overparameterized model.
- Suppose we observe (X_i, Y_i) $i = 1, \ldots, n$ with $X_i \in \mathbb{R}^d$. The linear model with least square can be formulated as the following optimization problem

$$\min_{\beta} f(\beta) = \frac{1}{n} \sum_{i=1}^{n} (X_i^T \beta - Y_i)^2.$$

- When $n \ll d$, the solution β^* is not unique and

$$f(\beta^*) = 0.$$

Overparameterized Model: Linear Regression

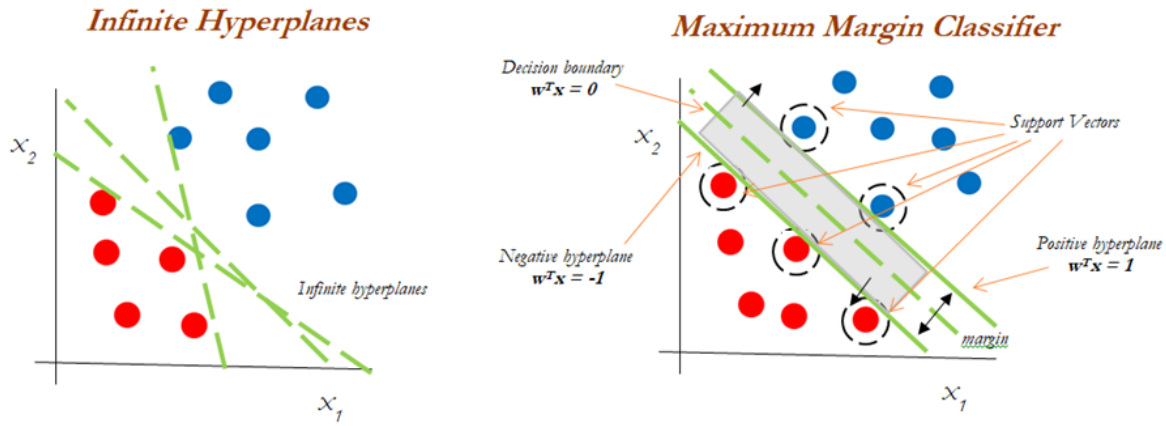
- If we choose gradient descent (or SGD), initialized at $\beta = 0$, we have a solution with minimizing L_2 norm

$$\min_{X\beta=Y} \|\beta\|_2$$

- If we choose coordinate descent, initialized at $\beta = 0$, we have a solution with minimizing L_1 norm

$$\min_{X\beta=Y} \|\beta\|_1$$

Overparameterized Model: Logistic Regression

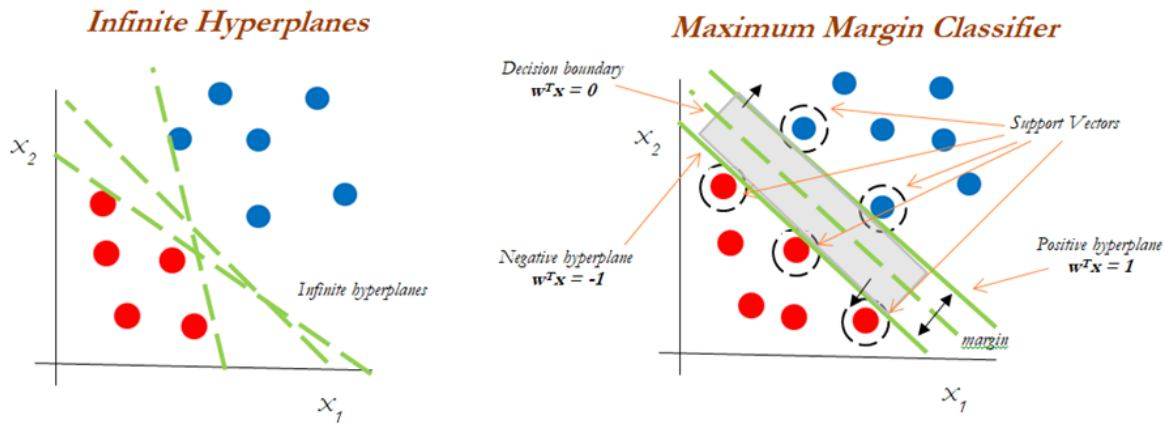


- Another example is classification when the data is linearly separable. We can consider the logistic regression

$$\min_{\beta} \frac{1}{n} \sum_{i=1}^{n} \ln\left(1 + \exp\left(-Y_i X_i^T \beta\right)\right).$$

- Clearly, the solution β^* is not unique.

Overparameterized Model: Logistic Regression



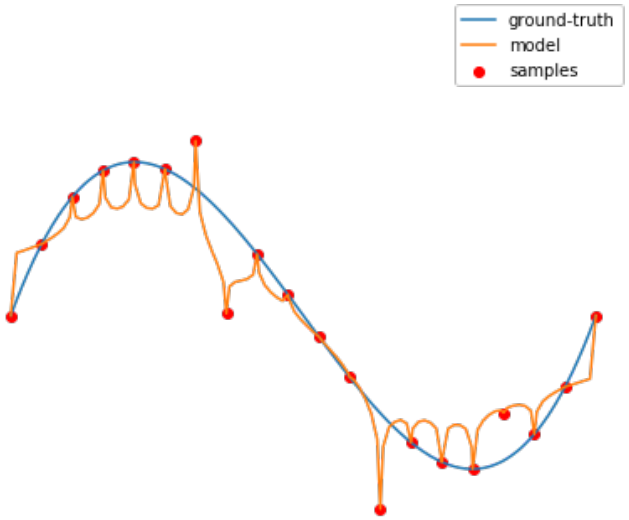
- If we apply gradient descent, we have

$$\frac{\beta^{(k)}}{\|\beta^{(k)}\|} \to \frac{\beta^*}{\|\beta^*\|},$$

where

$$\beta^* = \text{argmin} \|\beta\|_2 \qquad \text{s.t.} \quad Y_i X_i^T \beta \geq 1,\ i = 1, \ldots, n.$$

Overparameterized Model: Deep Neural Network

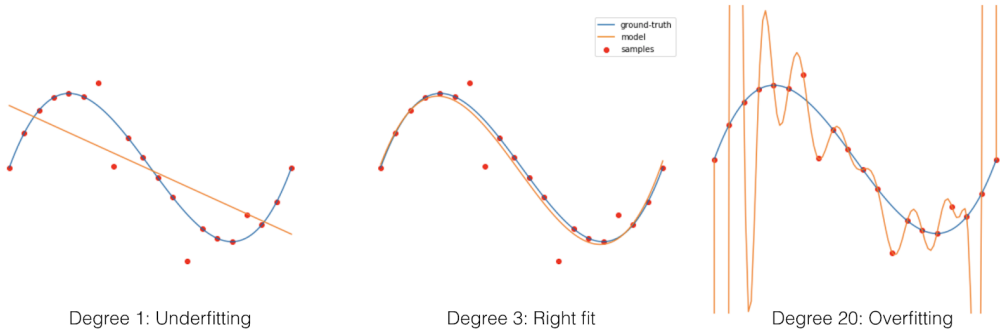


- In modern deep neural network, the model is super large and overparameterized. The resulting model is usually an interpolator, i.e.,

$$h_\beta(X_i) = Y_i, \qquad i = 1, \ldots, n$$

- The loss function can reach to zero by many interpolators.

Overparameterized Model: Deep Neural Network

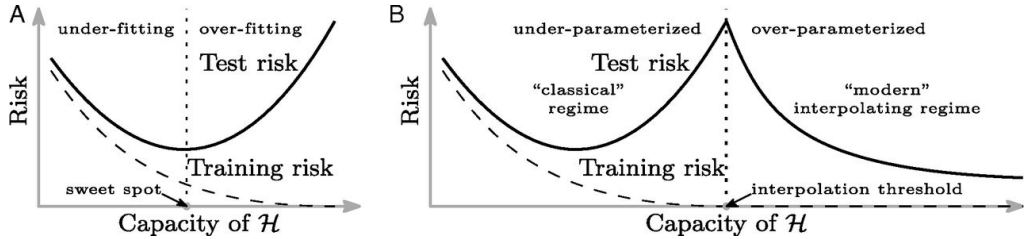


Degree 1: Underfitting Degree 3: Right fit Degree 20: Overfitting

- GD or SGD usually leads to a solution with minimum L_2 norm

$$\min_{\beta} \|h_\beta\|, \qquad \text{s.t.} \quad h_\beta(X_i) = Y_i, \qquad i = 1, \ldots, n$$

- So we can observe a double descent phenomenon.



Takeaway of Implicit Bias

- In classical statistics, we always design the loss function to make model selection. But we now realize that choosing optimization algorithm can also make model selection.
- When choosing optimization algorithm, we need to think about both convergence rate and implicit bias. Different implicit bias can lead to different resulting models and performance.
- Understanding optimization is now an important part of learning theory.